

Milestone 1 — Exploratory Data Analysis Report

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Dataset: Bank Customer Churn Prediction (Kaggle)

1. Dataset and Target

Dataset and Source: We chose the *Bank Customer Churn Prediction* dataset from Kaggle. The goal is to study customer attributes and see which ones relate to whether a customer leaves the bank. Link:

<https://www.kaggle.com/datasets/shantanudhakadd/bank-customer-churn-prediction>

Row Representation: Each row in the table represents a single bank customer. It includes their basic background (country, gender, age), account information (credit score, balance, number of products held), activity status, and whether they stayed or exited.

Dataset Size: The raw dataset contains **10,000 rows** and **14 columns**.

Target Column: The target column is **Exited**, which marks whether a customer churned or not. This is a binary classification target with 2 classes:

Target Class	Meaning	Count	Percentage
0	Customer stayed (retained)	7,963	79.63%
1	Customer left (exited)	2,037	20.37%
Total	All rows	10,000	100.00%

Input Features:

- **Identifier Columns (Dropped):** RowNumber, CustomerId, and Surname. These are IDs and customer names that don't help in predicting churn, so we dropped them.
- **Categorical Features (2):** Geography (France, Spain, Germany) and Gender (Female, Male).
- **Numerical Features (8):** CreditScore, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, and EstimatedSalary.

2. Initial Inspection and Data Quality

- **Duplicates:** We ran `df.duplicated().sum()` and found **0 duplicate rows**.
- **Missing Values:** We checked `df.isnull().sum()` across all columns and found **0 missing values** (the dataset is 100% complete).
- **Class Imbalance:** Around 80% stayed and 20% left. Because of this imbalance, accuracy alone won't be enough later on, and we will need precision and recall.
- **Unusual Observations:** Exactly 3,617 customers (about 36.2%) have a balance of 0, creating a sharp spike at zero in the balance distribution.

3. Descriptive Statistics for Numerical Features

Summary table for the 8 numerical features (\$N = 10,000\$), listing count, mean, standard deviation, min, median, and max.

Feature	Count	Mean	Std Dev	Min	Median	Max	Distribution
CreditScore	10,000	650.53	96.65	350.00	652.00	850.00	Bell-shaped
Age	10,000	38.92	10.49	18.00	37.00	92.00	Right-skewed
Tenure	10,000	5.01	2.89	0.00	5.00	10.00	Uniform (0-10)
Balance	10,000	76,485.89	62,397.41	0.00	97,198.54	250,898.09	Many zeros
NumOfProducts	10,000	1.53	0.58	1.00	1.00	4.00	Discrete (1-4)
HasCrCard	10,000	0.71	0.46	0.00	1.00	1.00	Binary (70.6% 1)
IsActiveMember	10,000	0.52	0.50	0.00	1.00	1.00	Binary (51.5% 1)
EstimatedSalary	10,000	100,090.24	57,510.50	11.58	100,193.92	199,992.48	Uniform

4. Target Distribution and Missing Values Plots

Here are the bar plots for target class balance and missing values check before any preprocessing.

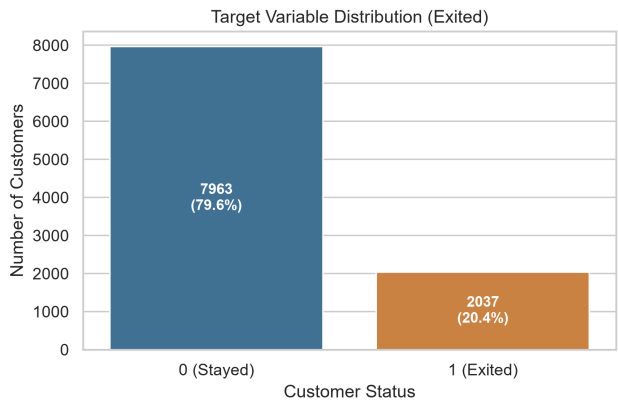


Figure 1: Counts for target column Exited.

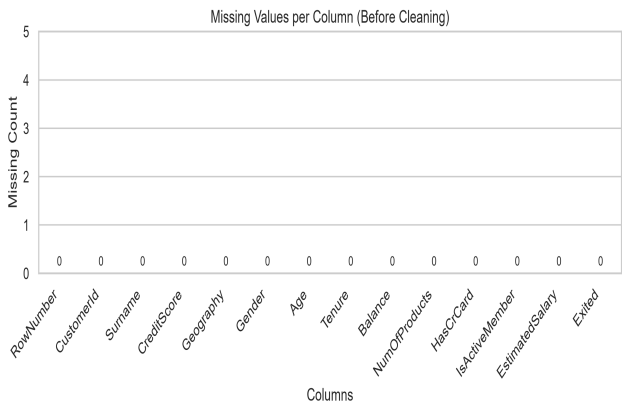


Figure 2: Missing values count per column.

Observations:

- **Target:** 7,963 customers stayed (79.6%) and 2,037 left (20.4%). The classes are imbalanced at roughly 4 to 1.
- **Missing Values:** None of the columns have missing entries (all counts are 0).

5. Estimated PDF for Numerical Features

We plotted normalized histograms (density=True) along with KDE curves for each of the 8 numerical features to see their shapes.

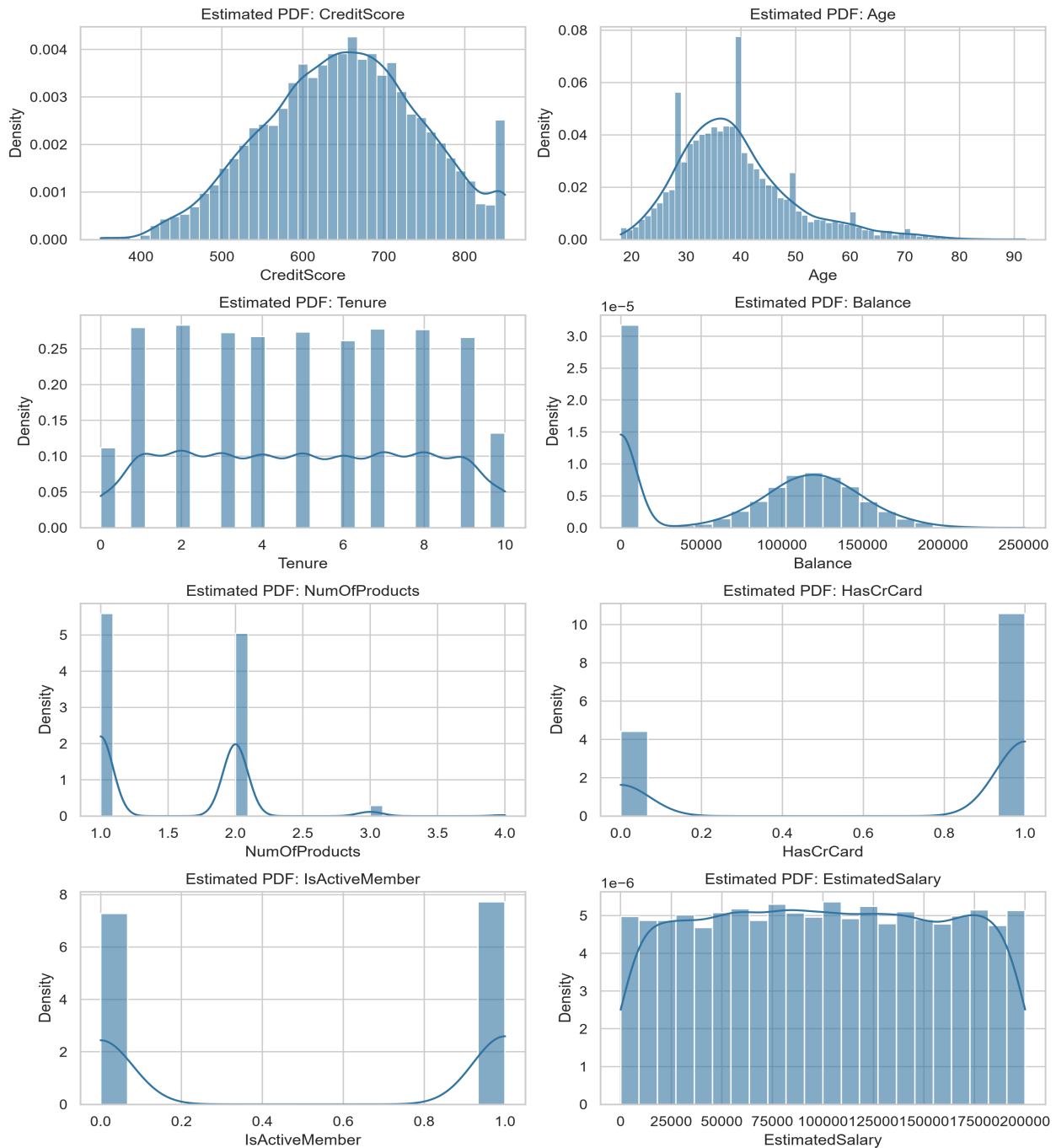


Figure 3: Estimated PDF histograms with KDE curves for numerical features.

Observations: CreditScore looks like a normal curve centered around 650. Age is skewed to the right, showing that most customers are in their 30s and 40s. Balance has a huge spike at 0 (people with no money in the account) and a second bump near 120,000. EstimatedSalary is flat from 0 to 200,000, which behaves like a uniform distribution.

6. Empirical CDF for Numerical Features

We plotted the empirical CDF step curves to inspect cumulative probabilities for each feature.

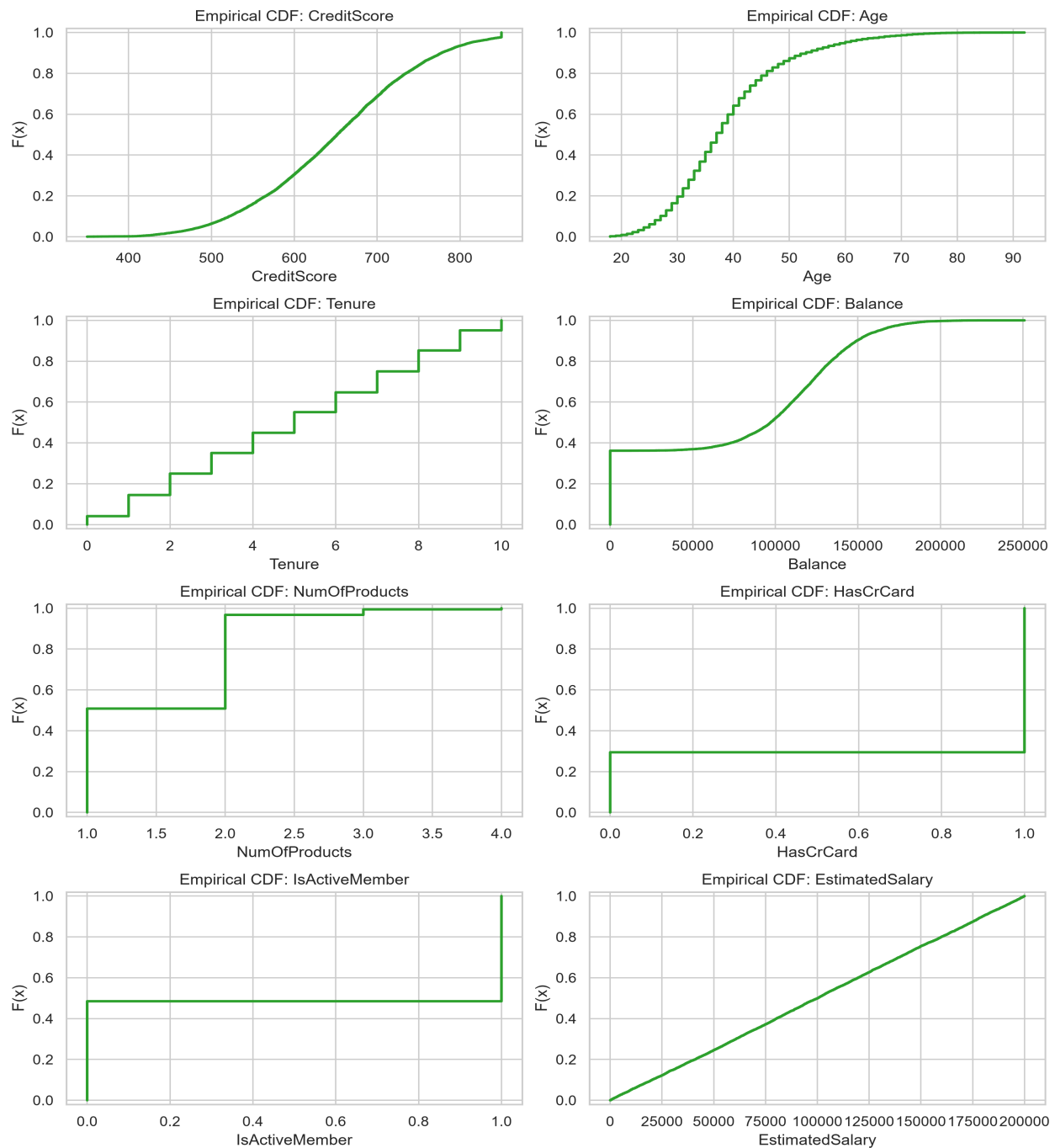


Figure 4: Empirical CDF step plots for numerical features.

Observations: The CDF for Balance jumps straight up at zero to about 0.36, confirming that about 36% of accounts have zero balance. EstimatedSalary shows an almost straight 45-degree diagonal line, which matches a uniform distribution. CreditScore and Age follow smooth S-shaped curves.

7. Correlation Matrix and Heatmap

Figure 5 shows the correlation matrix between numerical features. Figure 6 shows boxplots comparing Age and CreditScore for customers who stayed vs left.



Figure 5: Correlation heatmap of numerical features.

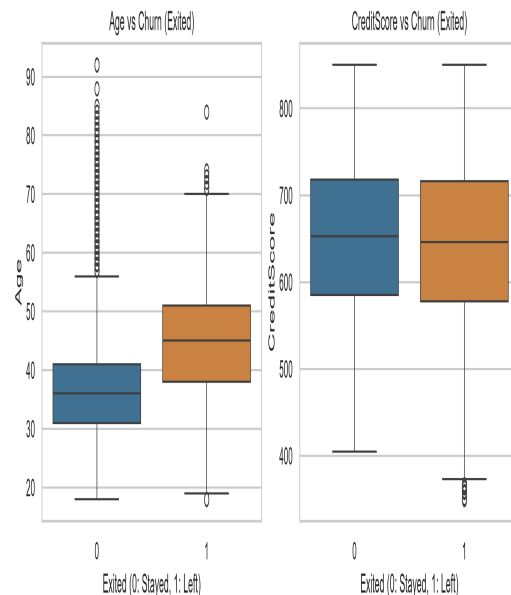


Figure 6: Boxplots of Age and CreditScore grouped by Exited.

Observations: Most numerical features have very low correlations with each other. The strongest correlation is -0.30 between Balance and NumOfProducts, meaning customers with more products tend to keep lower balances. In Figure 6, churned customers have a higher median age (~45) than retained customers (~36). Meanwhile, credit score values look practically the same for both groups.

8. Basic Preprocessing Summary

We followed the 4 required steps in order:

1. **Duplicate rows:** Checked with `drop_duplicates()`; 0 rows removed.
2. **Missing targets:** Checked with `dropna(subset=['Exited'])`; 0 rows removed.
3. **Missing numericals:** Checked for null values; 0 imputed because the columns had no missing entries.
4. **Encoding:** Dropped non-predictive IDs (RowNumber, CustomerId, Surname). One-hot encoded Geography into 3 binary columns (France, Germany, Spain) and Gender into 2 columns (Female, Male). Kept target Exited as integer labels (0 and 1).

9. Final Cleaned Dataset Specification

The cleaned dataset was saved to **Hloz_M1_Cleaned.csv**. It has **10,000 rows** and **14 columns**. Here is the column dictionary:

Column Name	Data Type	Transformation	Values / Range
CreditScore	Integer	Raw numerical	[350, 850]
Age	Integer	Raw numerical	[18, 92]
Tenure	Integer	Raw numerical	[0, 10]
Balance	Float	Raw numerical	[0.00, 250,898.09]
NumOfProducts	Integer	Raw numerical	[1, 4]
HasCrCard	Integer	Binary indicator	{0, 1}
IsActiveMember	Integer	Binary indicator	{0, 1}
EstimatedSalary	Float	Raw numerical	[11.58, 199,992.48]
Exited	Integer	Target (0: Stayed, 1: Left)	{0, 1}
Geography_France	Integer	One-Hot Encoded from Geography	{0, 1}
Geography_Germany	Integer	One-Hot Encoded from Geography	{0, 1}
Geography_Spain	Integer	One-Hot Encoded from Geography	{0, 1}
Gender_Female	Integer	One-Hot Encoded from Gender	{0, 1}
Gender_Male	Integer	One-Hot Encoded from Gender	{0, 1}

10. Conclusion and What We Learned

In this milestone, we explored the bank churn dataset of 10,000 customers. The data is very clean with no duplicate rows and no missing values in any feature. We found that the target variable Exited is imbalanced, where around 80% of customers stayed and 20% left. Looking at the distributions, CreditScore is approximately bell-shaped, EstimatedSalary is evenly spread across its range, and Balance has a noticeable peak at zero from customers with empty balances. From our correlation and boxplots, customer age has the clearest relationship with churn, as older customers tend to leave more often than younger ones. Finally, we removed unnecessary identification columns and one-hot encoded the categorical variables, leaving us with a clean table of 10,000 rows and 14 columns ready for modeling.